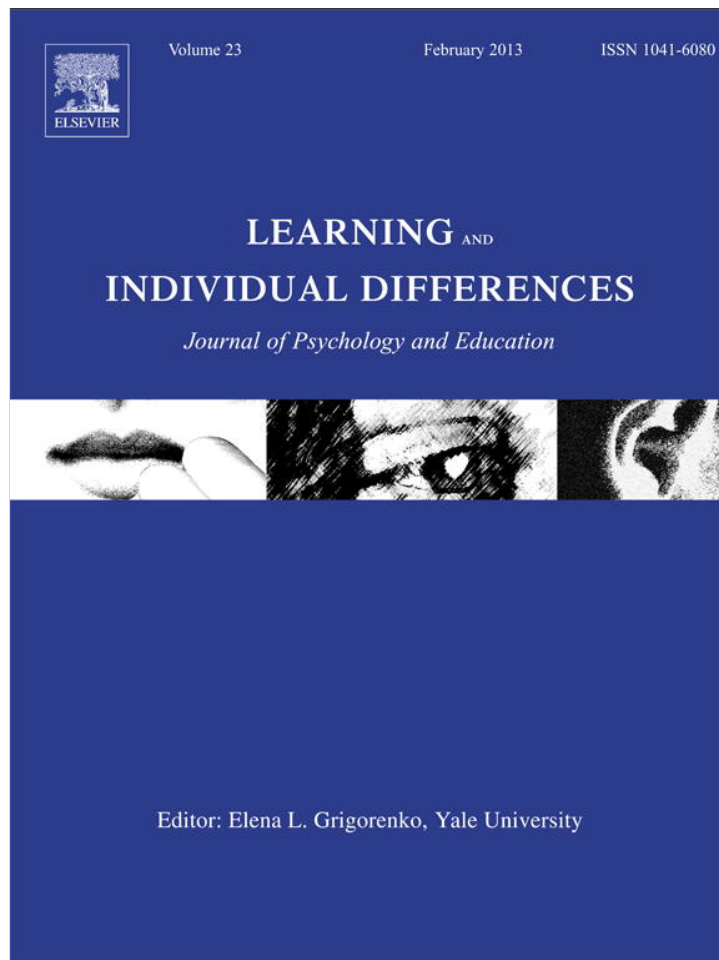




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Working memory capacity in preschool children contributes to the acquisition of school relevant precursor skills[☆]

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ABSTRACT

The aim of the present study was to investigate whether preschool children with limitations in the visual or phonological working memory are disadvantaged in the acquisition of school relevant precursor skills at school entry. A sample of 92 children was divided into three subgroups depending on their performance in visual and phonological working memory tasks in their last year before school entry (T_1). At T_1 the children were tested on cognitive measures as well as quantity–number competencies and phonological awareness tasks. Shortly after school entry (T_2), the precursor skills were assessed again. The findings indicate that preschool children with reduced phonological working memory capacity show weaker numerical competencies before school entry and weaker phonological awareness skills at the beginning of school compared to an unimpaired control group. On the other hand, visual working memory limitations in preschool only affect numerical competencies before and particularly after school entry.

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1. Introduction

Recently, the importance of specific precursor skills for successful learning in school has been demonstrated. Many studies emphasize the influence of phonological awareness on the acquisition of reading and spelling (Anthony et al., 2002; Bradley & Bryant, 1985; Goswami & Bryant, 1990; Kirby, Parrila, & Pfeiffer, 2003; Wagner & Torgesen, 1987) as well as the importance of early quantity–number competencies for the acquisition of arithmetic skills (Aunola, Leskinen, Lerkkanen, & Nurmi, 2004; Jordan, Kaplan, Locuniak, & Ramineni, 2007; Krajewski & Schneider, 2009). In addition to these early precursors of reading, spelling and arithmetic, recent studies indicate that domain-general cognitive capacities like working memory functioning are strongly associated with children's learning outcomes. For instance, it seems that working memory at school entry is closely associated to subsequent school achievement regarding literacy attainment (e.g., Gathercole, Brown, & Pickering, 2003) and math performance (e.g., Alloway & Alloway, 2010; Passolunghi & Siegel, 2001). However, research on the role of working memory for the acquisition of early domain-specific precursor skills is scarce. Therefore, the present study explores the impact of working memory

capacity on phonological awareness and early arithmetical precursors in young children.

1.1. Working memory

Working memory refers to the capacity to store and manipulate information over short periods of time. As Alloway (2009) points out, it reflects a relatively pure measure of a child's learning potential. According to the model provided by Baddeley (1986), working memory consists of three components, the phonological loop and the visual–spatial sketchpad as domain-specific, capacity-limited subsystems, and the central executive as a supervising modality-free component. The phonological loop is responsible for the temporary maintenance of verbal and auditory information. The visual–spatial sketchpad stores and processes static visual and spatial dynamic information. The central executive is responsible for focusing and switching attention as well as planning, monitoring, and regulating cognitive processes (Baddeley, 1996). Recent evidence suggests that these working memory components are in place at the age of 4 years (Alloway, Gathercole, & Pickering, 2006; Roebers & Zoelch, 2005).

1.2. Working memory and phonological awareness

The predictive value of central executive functioning (e.g., Alloway et al., 2005; Schuchardt, Kunze, Grube, & Hasselhorn, 2006) as well as phonological loop capacity on reading and spelling performance (e.g., Näslund & Schneider, 1996; Schneider & Näslund, 1993; Schuchardt et al., 2006; Wagner & Torgesen, 1987) has been

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demonstrated repeatedly. Furthermore, children with specific reading disabilities have deficiencies in phonological storage and processing (e.g., Pickering, 2006). However, much less is known about the impact of working memory on the emergence of precursor skills for reading and spelling such as phonological awareness, i.e. the awareness of the phonology of the spoken language (Wagner & Torgesen, 1987).

Phonological awareness tests are among the best predictors of early reading and spelling across different languages (cf., Adams, 1990; Bryant, MacLean, Bradley, & Crossland, 1990; De Jong & van der Leij, 1999; Näslund & Schneider, 1996; Torgesen, Wagner, & Rashotte, 1994). Although the relationship between phonological awareness and reading is undisputed, the underlying causality is discussed controversially (e.g., Castles & Coltheart, 2004). Some argue that phonological awareness precedes and facilitates reading acquisition (e.g., Bryant, Bradley, MacLean, & Crossland, 1989), while others argue that it develops alongside reading experience (e.g., Morais, Cary, Alegria, & Bertelson, 1979). Furthermore, there is still a debate about whether working memory and phonological awareness overlap (Bowey, 1996; Metsala, 1999) or represent separate entities (e.g., Gathercole, Willis, & Baddeley, 1991; Hecht, Torgesen, Wagner, & Rashotte, 2001; McDougall, Hulme, Ellis, & Monk, 1994). Alloway, Gathercole, Willis, and Adams (2004) argued that phonological awareness and working memory components are associated but separable constructs (e.g., De Jong & van der Leij, 1999). Nevertheless, several phonological awareness demands are also likely to rely on working memory resources. Particularly, different working memory capacities might facilitate the performance on phonological awareness skills differently. For instance, Näslund and Schneider (1991) found that phonological working memory capacity predicts phonological awareness in kindergarten. In addition to the phonological loop, central executive functioning seems to influence phonological awareness skills as well (e.g., Krajewski, Schneider, & Nieding, 2008; see also Alloway et al., 2005). However, taken together, research on the interplay of working memory capacity and phonological awareness, especially in very young children is still scarce and the relationship of these constructs requires further clarification.

1.3. Working memory and quantity–number competencies (QNC)

The acquisition of numerical competencies starts long before school entry. However, most of the literature addressing working memory and arithmetic concentrates on arithmetic achievement in school. In particular, central executive functioning was found to be related to math (e.g., Bull & Scerif, 2001; Geary & Hoard, 2001; Passolunghi, Vercelloni, & Schadee, 2007). Furthermore, a relationship between phonological loop capacity and math achievement was reported (Berg, 2008; Hecht et al., 2001; Towse & Houston-Price, 2001). Other studies suggest that phonological storage is related to mental calculation and the acquisition of number facts rather than to standard mathematic achievement tests (e.g., Adams & Hitch, 1997). Hitch and McAuley (1991) showed that children with arithmetical learning difficulties showed deficits in verbal short-term memory. Sometimes the visual–spatial sketchpad was also found to predict math achievement (Berg, 2008; Bull, Espy, & Wiebe, 2008; Holmes, Adams, & Hamilton, 2008; Rasmussen & Bisanz, 2005; Simmons, Singleton, & Horne, 2008), but not in all cases (e.g., Lee, Ng, Ng, & Lim, 2004).

So, the predictive power of different working memory components on math achievement differs across studies, depending on the children's age and the targeted math demands. Furthermore, the contribution of working memory to the acquisition of mathematical precursor skills as quantity–number competencies is far from clear. Krajewski and Schneider (2009) assume a growing influence of the visual–spatial working memory on numerical development, according to a recent theoretical model by Krajewski (2008). The model states that children pass through three developmental levels of arithmetical competence reflecting a growing understanding of the linkage between quantities

and number words. On the first level (QNC level 1) children acquire number words and the number word sequence without their semantic meaning. Here the number words are still isolated from quantities. On the second level (QNC level 2), children become aware that number words have a quantitative meaning and that numbers can be compared by size. They start linking number words to exact quantities more and more and know that the number–word sequence represents exact ascending quantities. Finally, at QNC level 3, children realize that number words represent not only quantities but also quantity relations. Krajewski and Schneider (2009) assume that to reach a deeper understanding of numbers at levels 2 and 3, i.e. for understanding the linkage between number words and quantities as well as quantity relations, visual–spatial working memory is especially important, while phonological awareness is relevant to the acquisition of the (verbal) number words at QNC level 1. Therefore, a well-functioning phonological working memory may support the acquisition of verbal arithmetical aspects like number facts in early childhood. So, working memory components seem to contribute differently in the course of early arithmetic development. In our study we wanted to shed further light on these relationships.

1.4. The current study

Although there is plenty of evidence that working memory is an important predictor of later school achievement, there is little research tapping the relation between working memory and school relevant precursor skills empirically. Thus, the major aim of the present study was to explore the impact of preschool working memory subcapacities on the acquisition of phonological awareness and quantity–number competencies before and shortly after school entry. In particular, three major hypotheses were examined:

1. Deficiencies in phonological working memory are related to low levels of phonological awareness.
2. Deficiencies in phonological working memory are also associated with poor early quantity–number competencies.
3. Deficiencies in visual working memory are related to poor early quantity–number competencies.

2. Method

2.1. Participants and procedure

92 children (47 girls) from kindergartens in Frankfurt/Main and surrounding areas were tested at two measurement points. Parental consent was obtained for each child. The first assessment was conducted in the last year of kindergarten (T_1). The children's average age was 68.3 months (SD 3.6, range 61–76). Each child was tested individually in a quiet room of the nursery school to obtain measures of working memory, nonverbal intelligence and object naming speed as well as early precursor skills of reading and arithmetic. Children were included in the sample based on their visual or phonological working memory performance. Children with working memory performance lower than one standard deviation from the mean standard score either in the visual or the phonological working memory domain were classified as having a deficit in the respective area. In detail, the first group contained children with low scores in the visual working memory task (T -score < 40) and an average performance in the phonological working memory task (at least half a standard deviation above the visual performance): visual deficit (VD); $n = 17$ (8 girls); 76.5% German as first language. The second group consisted of children with low scores in the phonological working memory task (T -score < 40) and an average visual working memory performance (at least half a standard deviation above the phonological performance): phonological deficit (PD); $n = 37$ (19 girls); 56.8% German as first language. The control group consisted of children achieving average or better scores (T -score ≥ 44) in both working memory tasks: control (C); $n = 38$ (20 girls); 57.9% German as first language. We used this

buffer to avoid that the control group includes children with a borderline working memory performance. Furthermore, the children in the control group were matched to both other groups according to age, gender, and native language. All children spoke German well enough to understand the test instructions.

One year later (T_2), the children were tested again on measures of precursor skills. This time the children were investigated in a room at their schools in groups of up to four children at most. At T_2 sample size dropped to 90 children because one child was not enrolled in school and another quit participation.

2.2. Measures

2.2.1. Working memory

Two tasks of the German computer-based working memory test battery for children from 5 to 12 years of age (AGTB 5–12; Hasselhorn et al., 2012) were used to assess phonological and visual working memory capacities. In the *word span forward task*, the child is asked to recall a sequence of one-syllable words in the presented order. In the *matrix span task*, the child is shown a pattern of black squares in a four by four matrix and has to recall this pattern by tapping the squares on a touch screen. Both tasks are adaptive, that is, test difficulty increases and decreases according to the child's performance. Reliability coefficients (Cronbach's α) for children from 5 to 8 years of age ($n = 726$) are .94 for word span forward and .98 for matrix span (see Hasselhorn et al., 2012).

2.2.2. Object naming speed

The rapid automatized naming task for objects assessed processing speed at which children were able to retrieve the names of objects from long-term memory (see Goswami, Schneider, & Scheurich, 1999; Swan & Goswami, 1997). The child is asked to name common objects (e.g., tree, ball, house) presented in two rows each with nine items as quickly as possible. Time was measured in seconds.

2.2.3. Nonverbal intelligence

To assess nonverbal intelligence, subtest 1 of a short German version of the Columbia Mental Maturity Scale (BUEVA; Esser, 2002) was used. The split-half reliability of children aged from 5.0 to 5.11 was .95 (Eggert, 1972).

2.2.4. Phonological awareness measures

Phonological awareness at T_1 was measured by a phoneme synthesis task. Following Lundberg, Frost, and Petersen (1988), in the *phoneme synthesis task* the child is presented a sequence of single phonemes and required to indicate which picture out of a set of four represents the appropriate word, corresponding to the presented phoneme sequence (maximum score = 8).

At T_2 , phonological awareness was measured by two tasks. The *phoneme analysis task* required the child to draw a cross for every sound he/she was able to differentiate in orally presented words. For instance, for the word “bus” the child had to draw three crosses, one for each sound of the word (b-u-s) (maximum score = 8). The *phoneme synthesis task* was similar to that one described at T_1 (maximum score = 8).

2.2.5. Quantity–number competencies (QNC)

Eight tasks were used at T_1 to measure QNC levels 1 and 2 (cf., Krajewski & Schneider, 2009). To assess level 1 competency, four tasks assessing children's handling of the number–word sequence were administered. In the first task, the child recites the number–word sequence from 1 to 31 (maximum score = 4). In the second task, the child is presented three number words and is then asked to name the subsequent number words (maximum score = 3). In the third task, the child is presented three number words and required to name the preceding number words (maximum score = 3). In the fourth

task, the child recites the number–word sequence from 5 backwards (maximum score = 2). Furthermore, a fifth task, the *Arabic–number task* was used. The child has to name randomly presented Arabic numbers from 1 to 20 (maximum score = 10).

To investigate level 2 competencies three tasks were employed. In the *quantity to number–word linkage task*, the child is asked to match quantities to the corresponding Arabic numbers and vice versa (maximum score = 5). In the *quantity to number–word linked seriation tasks* one item is missing in a sequence of items with ascending amounts of dots (from 1 to 8). The child has to select the correct item from a choice of five to fit into the row (maximum score = 3). In the next task, the child is asked to identify items that have fewer dots and items that have more dots than the child's age (maximum score = 4). In the *number–word comparison task*, the child compares pairs of verbally presented numbers and decides which is larger or smaller (maximum score = 8; cf., Krajewski & Schneider, 2009).

At T_2 , about three months (range 2.27–3.71) after school entry, seven tasks were used to assess all three QNC levels. These tasks were taken from a German test of basic mathematical competencies in first grade (MBK 1; Ennemoser, Krajewski, & Sinner, in preparation). Knowledge of the exact number word sequence (QNC level 1) was assessed with two tasks, *number dictation* and *number gap*. The first task assesses the child's ability to write ten dictated numbers in their Arabic form (maximum score = 5). The second task provides a row of two ascending numbers with one number missing in between. The child adds the number that fills the gap (maximum score = 4).

To assess the competence of linking number words with quantities (QNC level 2), two tasks were employed, *number line* and *number word comparison*. In the number line task the child is asked to assign numbers to horizontal or vertical number lines which range from 0 to 10. In one case the vertical number line ranges from 0 to 100 (maximum score = 5). In the *number comparison task*, the child compares two Arabic numbers by drawing a circle around the smaller one (maximum score = 4).

The linking of quantity relations with number words (QNC level 3) was assessed with three tasks: one more, one less, and number decomposition. In the first part of the *one more task*, pictures of pies containing a certain number of slices were presented. The child is asked to select the pie with one more slice than the target from a choice of four pies (maximum score = 2). In the second part of this task, only Arabic numbers are presented and the child specifies the number that contained “one more” (maximum score = 4). The *one less task* was similar to the two parts of the one more task apart from identifying “one less” than the target number (maximum score = 6). In the *number decomposition task*, the child solves basic addition tasks (maximum score = 4).

3. Results

Details describing the groups are shown in Table 1. For all working memory tasks span scores (raw data as well as age-specific T-scores) are reported.

The criterion for selecting the groups is reflected in *matrix span* and *word span forward*, which showed significant group differences in one way analyses of variance, $F(2, 89) = 53.48$, $p < .01$ and $F(2, 89) = 158.10$, $p < .01$, respectively. VD children performed significantly lower than the controls in the matrix span ($p < .001$). PD children showed significantly lower performance than the controls in the word span forward task ($p < .001$). Further analyses of variance showed no group differences with regard to age, $F(2, 89) = 0.56$, $p > .05$, performance on the object naming task, $F(2, 87) = 0.20$, $p > .05$ and intelligence, $F(2, 89) = 1.95$, $p > .05$. Because the standard scores of the BUEVA are obsolete, total scores of nonverbal intelligence are reported (maximum score = 57).

Table 2 summarizes descriptive data for the domain specific precursor tasks at T_1 (before school entry) and T_2 (after school entry). For all measures raw scores are reported, separated by groups.

Table 1
Descriptive statistics at T₁ (before school entry) separated by groups.

	Visual WM deficit (VD) <i>n</i> = 17		Phonological WM deficit (PD) <i>n</i> = 37		Control (C) <i>n</i> = 38	
	M	SD	M	SD	M	SD
Age (months)	67.82	3.56	68.78	3.82	68.05	3.52
Nonverbal intelligence (raw data)	48.53	4.93	48.97	5.58	50.97	4.89
Object naming speed (s)	22.40	6.45	23.68	7.05	23.10	7.31
Matrix span (raw data)	1.56	0.17	3.04	0.93	3.07	0.78
Matrix span (T-score)	35.29	3.00	54.19	8.51	55.18	6.59
Word span forward (raw data)	3.00	0.40	1.94	0.25	3.24	0.38
Word span forward (T-score)	53.00	6.61	34.32	3.58	56.97	6.87

Note. WM = working memory.

MANOVAs were performed on quantity–number competencies as well as on phonological awareness measures at both measurement points with group membership (VD, PD, C) as the fixed factor. Moreover, in pairwise comparisons, differences between VD and C as well as PD and C were tested using Dunnett's *T*-test ($p < .05$). This procedure allows for testing our three directional hypotheses. The hypotheses were that PD children perform lower on phonological awareness tasks and had poorer quantity–number competencies than the control children and that VD children had lower quantity–number competencies than control children. Due to the fact that means and standard deviations of the children's intelligence, naming speed, and age were comparable across the three groups, these variables were disregarded in further analyses.

3.1. Cross-sectional relationships

Results of univariate effects and pairwise comparisons regarding the relationships of working memory capacity and school relevant precursor skills in the last year before school entry (T₁) are reported in Table 3.

Analyzing early quantity–number competencies (QNC), group membership yielded significant univariate effects on levels 1 and 2. Pairwise comparisons demonstrated significant differences between the groups. VD and PD children scored significantly lower than the controls on all QNC measures at T₁.

Table 2
Descriptive statistics for quantity–number competencies and phonological awareness at T₁ and T₂ (maximum scores are shown in parentheses).

	Visual WM deficit (VD)		Phonological WM deficit (PD)		Control (C)	
	M	SD	M	SD	M	SD
Before school entry (T ₁)	<i>n</i> = 17		<i>n</i> = 37		<i>n</i> = 38	
QNC measures						
QNC level 1 (22)	12.59	5.79	11.60	5.31	15.64	4.62
QNC level 2 (20)	13.12	3.62	12.70	3.51	16.03	2.95
Total QNC score (42)	25.71	8.79	24.30	8.28	31.67	6.81
PA measure						
Phoneme synthesis (8)	5.00	1.97	4.30	1.82	4.74	2.18
After school entry (T ₂)	<i>n</i> = 16		<i>n</i> = 36		<i>n</i> = 38	
QNC measures						
QNC level 1 (9)	3.81	3.14	5.79	2.49	6.74	2.03
QNC level 2 (9)	5.31	2.15	5.83	1.98	6.61	1.78
QNC level 3 (16)	6.94	5.18	8.53	4.69	10.42	4.76
Total QNC score (34)	16.06	8.70	20.15	7.92	23.76	7.44
PA measures						
Phoneme analysis (8)	1.94	2.84	0.89	1.21	2.78	2.68
Phoneme synthesis (8)	5.75	1.73	5.89	1.65	6.33	1.81
Total PA score (16)	7.69	3.75	6.78	2.21	9.11	3.91

Note. WM = working memory; QNC = quantity–number competencies; PA = phonological awareness.

Concerning phonological awareness no differences between the groups were observed for the phoneme synthesis task.

3.2. Longitudinal relationships

Further MANOVAs were performed on QNC and phonological awareness tasks conducted three months after school entry (T₂). Univariate effects of group membership occurred for QNC levels 1 and 3 (see Table 4). Pairwise comparisons showed significant differences between VD and control children for all QNC levels. PD children still scored lower on all QNC levels than the controls, but this time the difference was not significant. With respect to phonological awareness a significant univariate group effect was found for the phoneme analysis. Pairwise comparisons identified significant differences between PD and controls. Like at T₁ no working memory effects were observed in the phoneme synthesis tasks.

4. Discussion

Working memory skills were shown to be associated with children's ability to learn, and with their later success in school (Gathercole et al., 2003). Moreover, a large body of research has focused on working memory impairments in children with learning difficulties in reading (e.g., Gathercole, Alloway, Willis, & Adams, 2006; Siegel & Ryan, 1989; Swanson, 2003) and mathematics (Bull & Scerif, 2001; Geary, Hoard, Byrd-Craven, & DeSoto, 2004; Gersten, Jordan, & Flojo, 2005). However, in these studies children are selected according to learning difficulties (representing the primary impairment), whereas working memory problems are secondary, associated characteristic. Therefore, little is known about the consequences of low working memory capacity, irrespective of other associated learning difficulties (Alloway, Gathercole, Kirkwood, & Elliott, 2009). To address this issue, the children of the present study were selected solely on the basis of low working memory capacity in visual or phonological storage.

Furthermore, little research has tapped the relationship between working memory and precursor skills. Thus, a major aim of the present study was to explore whether preschool working memory capacity is associated with early school relevant precursor skills. Before entering school children were classified as having a phonological (PD) or visual working memory deficit (VD). They were compared to a control group (C) of children with average working memory capacity on several quantity–number competencies and phonological awareness tasks, assessed before and shortly after school entry.

In line with the first hypothesis, results show that phonological loop deficiencies in preschool children influence phonological awareness performance. More precisely, in the longitudinal data, children with poor phonological storage capacity scored significantly lower than their matched controls in the phoneme analysis task. The related difference between PD and C children indicates that a child's capacity to store phonological information influences early phonological

Table 3
MANOVA results of quantity–number competencies and phonological awareness at T₁ with group membership as fixed factor.

Before school entry (T ₁)	Univariate effects			Pairwise comparisons	
	<i>F</i>	<i>p</i>	η^2	VD vs C <i>p</i>	PD vs C <i>p</i>
QNC measures					
QNC level 1	6.14	.003	.12	.041	.001
QNC level 2	10.45	.000	.19	.003	.000
PA measure					
Phoneme synthesis	0.85	.431	.02	.839	.284

Note. QNC = quantity–number competencies; PA = phonological awareness; C = control; VD = visual working memory deficit; PD = phonological working memory deficit.

Table 4

MANOVA results of quantity–number competencies and phonological awareness at T₂ with group membership as fixed factor.

After school entry (T ₂)	Univariate effects			Pairwise comparisons	
	<i>F</i>	<i>p</i>	η^2	VD vs C	PD vs C
				<i>p</i>	<i>p</i>
QNC measures					
QNC level 1	8.12	.001	.16	.000	.090
QNC level 2	2.97	.057	.06	.026	.081
QNC level 3	3.30	.041	.07	.016	.086
PA measures					
Phoneme analysis	6.44	.002	.13	.185	.001
Phoneme synthesis	0.88	.419	.02	.225	.235

Note. QNC = quantity–number competencies; PA = phonological awareness; C = control; VD = visual working memory deficit; PD = phonological working memory deficit.

awareness and corresponds to previous findings (see [Krajewski et al., 2008](#); [Näslund & Schneider, 1991](#)).

In contrast to our first hypothesis, no group differences were found for the phoneme synthesis tasks neither at T₁ nor at T₂. Different explanations may account for this result. First of all, the phoneme synthesis tasks did not seem to be very demanding for the children at both time points. Second, while phonological awareness tasks were often just presented verbally, in the phoneme synthesis task of this study the children received a sheet of paper displaying four pictures, one of which represented the target word. This visual support lowered phonological loop demands in order to create a pure phonological awareness task. However, we cannot rule out that the children identified the respective picture after the presentation of only a few phonemes, rather than the whole phoneme sequence. This might diminish variance in the task because demands both on sound blending and phonological working memory (i.e., storing a higher number of phonemes) were lowered. Nevertheless, our results show that phonological working memory deficits influence performance on tasks involving an analysis of phonemes in verbally presented words. Nonetheless, further research is needed, to evaluate the impact of working memory on a wider spectrum of phonological awareness skills.

Based on findings indicating a relationship between phonological loop capacity and math achievement in school (e.g., [Adams & Hitch, 1997](#); [Berg, 2008](#); [Hecht et al., 2001](#)) our second hypothesis suggested that deficits in the phonological working memory also lead to weaker performance in early arithmetical competencies. Results of T₁ confirmed this assumption. QNC levels 1 and 2 were significantly affected by phonological working memory capacity. Particularly, children with poor phonological loop capacity scored significantly lower than the control children on tasks of early quantity–number competencies assessed before school entry. This is in line with our assumption that in early childhood phonological working memory is relevant for the proper acquisition of the number–word sequence and Arabic numerals. Moreover, if the proper storage of phonological information like individual number words is necessary for learning the number–word sequence (QNC level 1), then it is not astonishing that children of the PD group with deficits on level 1 competencies encountered problems in the acquisition of level 2 competencies at T₁. However, the associations between phonological working memory deficits and arithmetical precursor skills changed from T₁ to T₂. So, in contrast to our assumptions, preschool phonological working memory deficits did not predict QNC at T₂. PD children still scored lower on these tasks, but there was no significant difference between these children and the controls. One possible explanation is that the influence of proper storage of verbal information is most relevant at the very beginning of acquiring the number word sequence.

The third hypothesis claimed that visual working memory deficiencies are associated with weaker early arithmetical precursor skills. As expected, VD children were outperformed by control children regarding all QNC levels at T₁ and T₂. The influence of visual–spatial

working memory on higher QNC levels corresponds to [Krajewski and Schneider \(2009\)](#), who claimed that visual–spatial working memory is particularly relevant to linking number words to quantities and quantity relations (cf., [Rasmussen & Bisanz, 2005](#)). However, our results also demonstrate an influence of visual working memory capacity on QNC level 1 at both measurement points. This contradicts the view that the visual–spatial sketchpad is not particularly relevant to the acquisition of the number–word sequence (e.g., [Krajewski & Schneider, 2009](#)). At this point, it is worth mentioning that the children participating in the study of [Krajewski and Schneider \(2009\)](#) were unselected, while the participants of the present study were selected according to low working memory capacity. Thus, this issue requires further research.

The associations between working memory components and arithmetical precursor skills changed from T₁ to T₂. We interpret this as a function of the different materials used to assess QNC at the two points in time. In kindergarten QNC level 1 was assessed by verbally asking and answering questions about number word sequences. At T₂ the children received small booklets containing several tasks which had to be solved in Arabic number form. Thus, the test material changed from a verbal to a visual presentation of each task. Specifically, in the first task of QNC level 1 at T₂ the child was asked to reproduce verbally presented number words by writing down each number. The reproduction of number words requires a visual representation format of those numbers.

In summary, the present results suggest that phonological loop capacity seems not only to be relevant to the acquisition of phonological awareness after school entry, but also to the acquisition of numerals and the number–word sequence before school entry. Additionally, visual working memory capacity seems to be relevant to preschool and later quantity–number competencies, in particular when school materials such as exercise sheets are used. So, we found evidence that working memory capacity contributes to school relevant precursor skills. While [Schuchardt et al. \(2006\)](#) stated that good phonological working memory resources are favorable conditions for the acquisition of reading, writing and mathematics, our findings add good visual working memory resources as another important prerequisite for the acquisition of arithmetic. In the past, there has been evidence that the dynamic component of the visual–spatial sketchpad influences math achievement (e.g., [Holmes & Adams, 2006](#)). Findings of our study now highlight that deficits in the static component, measured by matrix span, may also lead to weaker performances in early quantity–number competencies. Nonetheless, future investigations should preferentially assess the visual–spatial sketchpad using multiple tasks, operationalizing both the visual static and the dynamic spatial component (e.g., [Holmes et al., 2008](#); [Krajewski & Schneider, 2009](#)).

Certain limitations of the present study should be noted. The findings are based on a sample of children selected by their preschool working memory capacity. The control group was matched to the impaired groups by age, gender, and native language. Although these groups did not differ in terms of intelligence and naming speed, social factors, such as socio-economic background or parental level of education, were not considered. Furthermore, sample size, especially the one of the VD group, was very small. Moreover, neither the children's duration in nursery school nor the educational concepts of each nursery school were taken into account. Even though informal schooling in German kindergartens is not common, we cannot rule out the possibility that it took place in some cases.

5. Conclusions

These results have practical implications for daily nursery school life. A combination of knowledge-based and cognitive measures, such as working memory, may lead to a good estimation of a child's learning potential and a proper diagnosis can provide the basis for identifying children at risk of developing learning disabilities at an early stage in their lives. In turn, such an early identification helps in responding to

the special demands of these children effectively. This may for instance include adapting learning situations to children's limited working memory resources (cf., Gathercole, Lamont, & Alloway, 2006). The teacher's awareness of working memory limitations in children may enhance the deliberate use of strategies and methods that reduce working memory demands in learning arrangements. In turn, this may enable children with limited working memory resources to acquire important school relevant precursor skills such as early quantity–number competencies and phonological awareness skills resulting in better school achievement.

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